



MVP Title

Distributed MPI Job Execution on Kubernetes (Mini HPC Cluster)



MVP Goal (Keep it Focused)



Run a parallel HPC job (MPI) across multiple Kubernetes pods



Show scaling + execution + output collection

That's it. No overengineering.



MVP Architecture (Simple but Powerful)

User → kubectl → Kubernetes Cluster → MPI Pods → Shared Storage → Output

Components you'll use:

Kubernetes (single master + 2 workers)

Docker (for MPI image)

MPI (OpenMPI)

NFS (shared volume)



Step-by-Step MVP Plan

- ◆ Step 1: Setup Cluster (Local or Cloud)

Option A (easy):

3 VMs using VirtualBox (like your CDAC setup)

Option B:

AWS EC2 (1 master + 2 worker nodes)

Install using:

kubeadm

- ◆ Step 2: Create MPI Docker Image

Dockerfile

Dockerfile

FROM ubuntu:20.04

RUN apt update && apt install -y \

openmpi-bin openmpi-common libopenmpi-dev python3

WORKDIR /app

COPY matrix.py .

CMD ["mpirun", "-np", "4", "python3", "matrix.py"]

- ◆ Step 3: Sample HPC Job (Parallel Code)

Python

matrix.py

from mpi4py import MPI

comm = MPI.COMM_WORLD

rank = comm.Get_rank()

print(f'Hello from process {rank}')

- ◆ Step 4: Kubernetes Job YAML

YAML

```
apiVersion: batch/v1
```

```
kind: Job
```

```
metadata:
```

```
  name: mpi-job
```

```
spec:
```

```
  template:
```

```
    spec:
```

```
      containers:
```

```
      - name: mpi-container
```

```
        image: your-dockerhub/mpi-app
```

```
        restartPolicy: Never
```

♦ Step 5: Deploy Job

Bash

```
kubectl apply -f mpi-job.yaml
```

```
kubectl get pods
```

```
kubectl logs <pod-name>
```

👉 Output will show:

```
Hello from process 0
```

```
Hello from process 1
```

```
Hello from process 2
```

```
Hello from process 3
```

💥 This proves parallel HPC execution

♦ Step 6: Add Scaling (Very Important)

Edit job:

YAML

```
parallelism: 3
```

```
completions: 3
```

👉 Now multiple pods run simultaneously → mini HPC cluster behavior

♦ Step 7: Add Shared Storage (Optional MVP+)

Use:

NFS

👉 Store outputs from all nodes in one place

👤 Demo Flow (CRUCIAL 🚨)

Show cluster nodes

Bash

```
kubectl get nodes
```

Deploy MPI job

Show multiple pods running

Show logs (parallel output)

Increase parallelism → redeploy

Show faster / scaled execution

📊 MVP Features (What You'll Claim)

- ✓ Distributed execution
- ✓ Parallel processing (MPI)
- ✓ Containerized HPC
- ✓ Kubernetes job orchestration
- ✓ Basic scaling

🧠 What Makes This MVP Strong?

Even though it's small, it demonstrates:

HPC fundamentals ✓

Cloud-native architecture ✓

Parallel computing ✓

Real-world scalability ✓

👉 90% students won't even reach this level properly.

🔥 MVP → Upgrade Path (if time permits)

Add just ONE of these:

1. 📊 Monitoring

Prometheus + Grafana

2. 🌐 Simple UI

Flask API to submit jobs

3. ⚡ Smart Scheduler

Volcano Scheduler